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Research Interests

learning-to-rank, world models, large language models, multi-agent systems, and scalable ML systems.

Education

Expected Jul. 2026 **MSc Computer Science, University of Amsterdam, Netherlands**

○ GPA: **8.04**

○ Relevant coursework: Deep Learning 1 (8.5), Deep Learning 2 (8.0), Programming Large-Scale Parallel Systems (9.5), Web Data Processing Systems (8.5)

Jul. 2020 **B.Tech. Computer Science, PES University, Bangalore, India**

○ GPA: **8.24/10**; graduated First Class with Distinction.

○ Relevant coursework: Machine Learning, Data Structures, Social Network Analytics, Big Data, and Data Analytics.

Master's Thesis

Jul. 2026 **Smart Rankings, Qogita, Amsterdam**

○ Built an end-to-end learning-to-rank pipeline over **52M+ product impressions, 163k buyers, and 450k products** using LightGBM LambdaMART.

○ Engineered **78** point-in-time ranking features spanning behavioural history, semantic embeddings, buyer-product affinity, and session context.

○ Improved production ranking performance by **9.55 percentage points in NDCG@10** through empirical label-gain optimisation and hyperparameter tuning.

○ Developed reliable large-scale data pipelines, offline evaluation infrastructure, and feature-engineering workflows to serve production-scale traffic.

Research Projects

Jun. 2026 **Reward-Aware World Models Using Joint-Embedding Predictive Architectures (JEPA)**

○ Investigated whether latent representations learned by Joint-Embedding Predictive Architectures encode recoverable reward and value information for downstream reinforcement learning.

○ Implemented reward and value probing, reward-aware fine-tuning, and latent-space model predictive control across multiple Gymnasium environments.

○ Compared JEPA world models with DreamerV3 and DINOv2 baselines, evaluating reward-prediction quality, latent planning, and representation learning.

○ Showed that reward supervision is broadly compatible with the JEPA objective: jointly optimising reward/value, latent-prediction, and SIGReg losses stabilises fine-tuning across all 8 environments without representation collapse.

May 2026 **Coordination Without Commitment: Multi-Agent LLM Coordination**

○ Investigated coordination failures of decentralised large language model agents on distributed computing tasks, including consensus, graph colouring, leader election, matching, and vertex cover.

○ Designed the Neighbour Belief Protocol (NBP), a structured belief-sharing protocol that improves the coordination of small language models on perception-limited tasks.

○ Conducted extensive benchmarking across communication graphs, model scales, and coordination protocols, identifying distinct failure modes in multi-agent reasoning.

○ Demonstrated that protocol-level improvements can substantially reduce the gap between smaller open-weight models and larger frontier models.

- Sep. 2025 **LLM-Assisted Hindi Information Extraction for DBpedia**
- Developed a hybrid multilingual information-extraction system combining rule-based NLP with small language models, achieving **66% recall** on the Hindi-BenchIE benchmark.
 - Built a complete knowledge-graph population pipeline, including entity linking, predicate linking, ontology alignment, and SPARQL endpoint deployment.
 - Published at the **ESWC 2026 Text2KG Workshop**
- Jun. 2025 **Parallel K-Means Using CUDA, *Programming Large-Scale Parallel Systems***
- Implemented a GPU-accelerated K-Means clustering algorithm in C++/CUDA.
 - Achieved up to a **47.6× speedup** over the sequential CPU implementation through parallel distance computation and optimised GPU kernels.
 - Performed scalability experiments on the DAS-5 supercomputer across varying dataset sizes, dimensionalities, and cluster counts.

Industry Experience

- May 2025–Sep. 2025 **Google Summer of Code Contributor, *DBpedia Association***
- Competitively selected from a global applicant pool to contribute to DBpedia’s multilingual knowledge-extraction infrastructure.
 - Designed multilingual knowledge-extraction pipelines for Hindi Wikipedia using small language models and rule-based NLP.
 - Developed neural information extraction, predicate linking, ontology alignment, and knowledge-graph enrichment pipelines integrated into DBpedia.
 - Contributed production code to the DBpedia Information Extraction Framework and the Hindi DBpedia chapter.
- Oct. 2020–Jul. 2024 **Senior Software Engineer, *Lendingkart*, Bangalore, India**
- Redesigned a production decision-rule engine for real-time credit scoring migrating from a monolithic Python service to an event-driven Java architecture achieving a **99.4% application success rate** and reducing end-to-end latency from ~60 s to ~10 s.
 - Built an event-driven loan reconciliation system processing high-volume repayment streams and reconciling them against customer accounts in real time, eliminating 3 man-days/week of manual work and removing a class of reconciliation errors entirely.
 - Optimised a multi-tenant real-time analytics platform on the ELK stack with tenant-scoped views and RBAC, cutting dashboard load latency by ~40% and establishing monitoring that reduced MTTR by ~15% for critical services.
 - Built an API integration platform reducing partner onboarding from 2–3 dev-days to under 30 minutes per API, enabling self-serve workflows for non-technical users and accelerating onboarding of **30+ external partners**.

Publications

- 2026 **LLM-Assisted Multilingual Information Extraction for Knowledge Graph Population: The DBpedia Hindi Chapter.** Accepted at the **ESWC 2026 Text2KG Workshop**.
- 2021 **A Framework to Capture the Shift in Dynamics of a Multi-phase Protest.** Springer Communications in Computer and Information Science (CCIS).

Technical Skills

Languages Python, Java, SQL, Bash, SPARQL

Machine Learning PyTorch, LightGBM, Transformers, large language models, learning-to-rank, information retrieval, knowledge graphs

Infrastructure Airflow, DBT, Snowflake, Docker, Kubernetes, Kafka, Elasticsearch, Redis, Git